15.1 Optimization, Part 1

Let $z = f(x,y)$ be a function. Just like in calc I:

- f has a **local maximum** at (a,b) if $f(a,b) \geq f(x,y)$ for all (x,y) near (a,b).
- f has a **local minimum** at (a,b) if $f(a,b) \leq f(x,y)$ for all (x,y) near (a,b).

How do we find local extrema (i.e., local max's and min's)?

⇝ A point (a,b) in the domain of f is called a **critical point** of f if

- $\nabla f(a,b)$ is undefined, or
- $\nabla f(a,b) = \langle 0,0 \rangle$.

Then if f has a local max or min at (a,b), then (a,b) is a critical point.

<u>Warning</u> Not all critical points are locations of extrema! If $\nabla f(a,b) = \langle 0,0 \rangle$ but there is no max or min at (a,b), then f has a **saddle point** at (a,b).

→ Do problems 1–4.

Second Derivative Test

Suppose $\nabla f(a,b) = \langle 0,0 \rangle$ and set

$$D = \begin{vmatrix} f_{xx}(a,b) & f_{xy}(a,b) \\ f_{xy}(a,b) & f_{yy}(a,b) \end{vmatrix} = f_{xx}(a,b) f_{yy}(a,b) - f_{xy}(a,b)^2$$

(1) If $D > 0$ and $f_{xx}(a,b) > 0$, then f has a local min at (a,b).

(2) If $D > 0$ and $f_{xx}(a,b) < 0$, then f has a local max at (a,b).

(3) If $D < 0$, then f has a saddle point at (a,b).

(4) If $D = 0$, the test is inconclusive.

→ Do problems 5 and 6.